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S510/S520 – Master Seminar on Econometrics

Google Search Indicators and Nowcasting German Unemployment: Early vs. Late Information

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Submission date:	January 9, 2026

Abstract

This paper builds on Zimmerman’s approach to evaluate whether biweekly Google search indicators can improve one-step-ahead nowcasts of German unemployment. Weekly search series from Indeed, Stepstone, Jobbörse, Arbeitsamt, and Bewerbung are combined into consistent 2011–2023 histories and aggregated into biweekly components reflecting the W34 (M1) and W12 (M) schemes, in line with Zimmerman’s framework. In contrast to the original approach, several pre-tests are conducted, including the Augmented Dickey–Fuller and Engle–Granger cointegration tests. Trend indicators are then modeled within an error-correction framework. Finally, early versus late nowcasts are compared, and predictive gains are assessed using out-of-sample RMSE, MAE, and Diebold–Mariano tests. The analysis compares both window specifications to identify which provides more informative signals for real-time labor market assessment.

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1 Introduction

Central banks, research institutions, and economic policymakers rely on timely indicators to monitor macroeconomic conditions and to evaluate the stance of fiscal and monetary policy. However key statistics such as unemployment, consumer spending, or inflation, are released only with a considerable delay after the reference period. These publication delays create information gaps especially when rapid responses are needed, for instance during economic crises, sudden demand shocks, or turning points in the business cycle. The COVID-19 pandemic and the 2008 financial crisis vividly demonstrated how delays in labour and consumption statistics hinder timely policy responses.

The instant and timely knowledge of market developments or business sentiment is hence crucial for authorities seeking to anticipate rather than react to economic shifts. Nevertheless, data collection processes especially based on surveys or administrative records are slow and often resource intensive. The result of that is that official indicators often fail to capture real-time economic movements.

With regard to this situation, researchers and institutions more and more explore the potential of alternative, high-frequency data sources, that can complement traditional statistics and enhance short-term and even maybe long-term forecasting. Examples here entail digital traces from online job postings, mobility records, credit card transactions, and, most prominently, search engine activity. Each of these reflects aspects of real-world behavior and expectations in near real time.

Among these new data sources search engine queries the ones from google trends stand out, as they publicly available and high-frequent. Changes in search intensity of specific words such as “jobs”, “application”, or “recession” may reveal shifts in labor-market sentiment or employment expectations long before those trends appear in official statistics.

Search engines have become the primary gateway to information on the internet, enabling users to locate and filter vast amounts of content. They democratize access to information across social and demographic groups and are used by virtually all segments of the population for a variety of purposes, including job search, product research, and education. Consequently, search engines generate massive volumes of data on what individuals are currently interested in, concerned about, or intend to do next. Google’s dominance in the search market makes its data particularly representative. As of September 2025, Google accounts for approximately 85 % of desktop and over 92 % of mobile search engine usage in Germany, ensuring that Google Trends provides broad and demographically diverse coverage of online search behavior.

The theoretical appeal of search data lies in their behavioral immediacy. Individuals’ online searches often precede actual economic decisions. When people become concerned about job security or begin seeking employment, they start searching for relevant information online—well before these changes are captured in official labor statistics. Similarly, searches

for terms like “recession” or “inflation” may reflect emerging shifts in economic sentiment. Aggregated across millions of users, these search activities form a type of micro-level behavioral data that can serve as a high-frequency proxy for macro-level dynamics. In this sense, search activity acts as a digital thermometer of the economy, translating intentions and expectations into measurable signals. Because of this timeliness and behavioral interpretability, search data are particularly suited for nowcasting especially for indicators such as unemployment that are released with delay.

One of the earliest contributions to this line of research comes from **ettrege**, who investigated the relationship between web-based search data and U.S. unemployment statistics even before Google Trends was available. Their study used data from WordTracker’s metasearch engine reports to examine whether the frequency of job-related search terms could anticipate subsequent movements in unemployment. They found that while short-term search fluctuations carried limited predictive content, longer-term averages of search activity exhibited strong associations with future unemployment—sometimes outperforming traditional weekly claims data. This early work established the fundamental intuition that aggregated online behavior can contain predictive signals about labor market conditions.

This concept was advanced by **Askitas2009** that introduced the google predictor which was a systematic attempt to quantify the informational content of Google search data for macroeconomic forecasting. Using German data from 2004–2009, they demonstrated that job-related search terms correlate strongly with subsequent unemployment figures published by the Federal Employment Agency. Their error-correction models showed that increases in such searches reliably anticipated rises in unemployment weeks before official data releases, confirming that online search behavior often precedes observable labor-market movements.

Building on this **varian** extended the use of Google search data to a broader range of macroeconomic indicators. Rather than traditional forecasting, their approach emphasized nowcasting, estimating indicators that are available only with delay by using contemporaneous search data. They demonstrated that incorporating search query volumes, such as “unemployment benefits” or “car sales”, into autoregressive models significantly improved out-of-sample performance, particularly around turning points in the business cycle. Their work established Google Trends as a practical and publicly accessible data source for real-time economic monitoring.

Subsequent studies such as **carrier**, examined whether these results also hold in emerging markets. Focusing on Chilean automobile sales, they constructed a Google Trends Automotive Index and showed that including search data improved both in-sample and out-of-sample forecasts and better captured market turning points. Their analysis also confirmed that search activity contains information that precedes official data, not merely reacts to it.

D’Amuri2010Google refined these ideas further by constructing a Google Job Search Index (GI) for the U.S. They compared more than 500 forecasting models which include several linear and nonlinear ARMA and ARMAX specifications, to test whether Google search data improved predictions relative to traditional indicators such as initial unemployment claims. Using extensive out-of-sample evaluation (Diebold–Mariano tests, forecast encompassing, White’s Reality Check), they found that models including the GI consistently outperformed benchmarks in over 70 % of U.S. states. Moreover, quarterly forecasts based on their models performed comparably or better than those from the Survey of Professional Forecasters (SPF). Their findings strongly supported the notion that online search activity contains timely and leading information about labor-market dynamics.

While most early research focused on macroeconomic indicators such as unemployment or inflation, **goel** expanded the focus to consumer behavior, using Yahoo! search data to predict entertainment product demand. They found that search intensity was highly correlated with future sales and popularity, often outperforming traditional predictors when baseline data were scarce. This work emphasized that web search data’s predictive advantage depends on the context—it is particularly valuable for new or fast-changing markets where official data are limited.

Complementary research also explored diverse areas: Guzman (2011) on inflation, Baker and Fradkin (2011) on job-search responses to policy changes, and Preis et al. (2010) and Schmidt & Vosen (2009) on consumer sentiment and retail sales. These studies collectively underscore the versatility of search data as a real-time barometer of social and economic activity.

Despite extensive international evidence, relatively few recent studies have revisited the German labor market using up-to-date Google Trends data. Given Germany’s economic size, its distinct labor-market institutions, and the increasing penetration of digital search behavior (with over 85 % of searches conducted via Google), re-evaluating the predictive power of search data in this context remains of high relevance.

Against this background, the present seminar paper follows the general framework of **Zimmermann2012** and investigates whether search-intensity data from Google Trends can improve the nowcasting and short-term forecasting of German unemployment. The central idea is that search terms related to job seeking or applications may contain leading information about changes in the unemployment rate, reflecting individuals’ job-search behaviour even before official labour statistics are released. To examine this relationship, I construct a monthly panel covering the period from mid-2011 to January 2023, combining official unemployment data from the Federal Employment Agency with bi-weekly Google Trends indicators for several search terms. This period offers both sufficient length for robust econometric estimation and enough variation to capture major economic phases, including the recovery from the Eurozone crisis, the COVID-19 pandemic, and the subse-

quent inflationary period. Because Google Trends provides only five years of weekly data per download, the series are carefully stitched from overlapping time windows to obtain a continuous historical record. Following **Zimmermann2012**, the analysis distinguishes between two bi-weekly windows, where one covering the two weeks prior to the official unemployment release and another capturing the two weeks after. This is done in order to test whether search behaviour anticipates or reacts to labour-market outcomes. Before estimating the models, each series is examined for stationarity using Augmented Dickey–Fuller tests and inspected for potential deterministic trends and seasonal components, which are common in labour-market indicators. To investigate the long-run equilibrium relationship between unemployment and search activity, I employ the Johansen cointegration framework and estimate a set of Vector Error Correction Models (VECM). Each VECM includes the unemployment rate and the corresponding two Google Trends variables, allowing both short and long-run dynamics to be jointly identified. The cointegration rank is determined via the Johansen trace and maximum-eigenvalue tests. Once the VECMs are estimated, I evaluate their in-sample fit through nowcasting exercises, replicating Zimmermann’s approach by reconstructing the in-sample unemployment path from the fitted differences. In addition, I extend the analysis by performing out-of-sample forecasting with both expanding and rolling estimation windows. This enables a comparison of the models’ predictive accuracy across horizons of one, two, and three months, going beyond the scope of earlier studies such as **Zimmermann2012** and **Askitas2009**. Finally, impulse-response analyses provide further insight into the short- and long-term reactions of unemployment to shocks in search intensity, helping to assess the behavioural and policy relevance of these relationships. The next section introduces the data in greater detail, outlining the sources, transformations, and construction of the Google Trends variables and official unemployment series used in the empirical analysis.

2 Data

2.1 Official unemployment series (Germany)

The target variable in this study is the *seasonally unadjusted* monthly unemployment rate for Germany published by the *Bundesagentur für Arbeit* (Federal Employment Agency). Concretely, I use the nationwide series labelled (Arbeitslosenquote bezogen auf alle zivilen Erwerbspersonen), i.e., the unemployment rate relative to the entire civilian labour force. The series is expressed in percentage points at monthly frequency. Following **Zimmermann2012**, I deliberately work with the unadjusted rate rather than a seasonally adjusted counterpart. This choice preserves the raw cyclical and calendar-related variation that is informative for high-frequency indicators and is standard in the related literature on search-based nowcasting for Germany. To maintain consistency with prior work,

I also exclude alternative denominators (e.g. “abhängige zivile Erwerbspersonen”) and regional disaggregations (West/East); only the all-Germany aggregate is retained. For empirical transparency, I use a single, latest-available vintage downloaded from the Federal Employment Agency’s statistical releases. In the next subsection I motivate the sample window used in the analysis and describe the exact transformations applied to construct the modelling panel.

In contrast to the paper of **Zimmermann2012**, where the authors relied on a relatively short observation window, covering January 2004 to April 2009. This study employs a substantially longer and more recent sample spanning mid-2011 to January 2023. The decision to focus on this timeframe is motivated by both data-availability constraints and economic considerations.

From a practical point of view, the choice to begin the sample in 2011 is motivated by both data-related and economic considerations. On the data side, Google Trends data are officially available from 2004 onward, but the early years of the platform are known to be less reliable. Before 2011, Google’s underlying systems for query collection and normalization were repeatedly revised, and the earlier “Google Insights” interface used in Zimmermann’s study (2009) was later replaced by the current Google Trends format. These changes bring difficulty when it comes to ensure that search volumes from the mid-2000s are directly comparable with those of more recent years. Moreover, internet penetration in Germany increased sharply over that decade, meaning that search activity before 2010 may not yet have represented the broader population. By starting in 2011, the analysis builds on a period where Google search behavior had become widespread, data were recorded in a consistent format, and the search indices show stable, reproducible patterns across downloads.

At the same time, the choice of 2011 as a starting point is also justified from the perspective of the unemployment series itself. During the mid-2000s, German unemployment was shaped by the aftermath of the Hartz reforms and the recovery from the early-2000s recession. By 2010–2011, these structural adjustments had largely played out, and the labor market entered a phase of relative stability and sustained improvement. Beginning the analysis at this point therefore avoids structural breaks linked to policy reforms, providing a cleaner relationship between online job-search behavior and the observed unemployment rate. It also creates a more symmetric window that includes both expansionary and contractionary phases. It entails the post-Eurozone crisis recovery, through the low-unemployment years before COVID-19, to the pandemic shock and the inflationary period after 2022.

Of course, this decision also has trade-offs. Excluding the earlier years reduces the historical depth of the sample and means that the results cannot directly replicate Zimmermann’s 2004–2009 findings. However, the improved data quality, stable coverage, and economic relevance of the 2011–2023 period arguably outweigh these drawbacks. The resulting

dataset captures more than a decade of modern labor-market dynamics in Germany, offering both a longer and more diverse testing ground for assessing whether search activity continues to anticipate changes in unemployment.

2.2 Google trends data

Google Trends is an analytical tool that provides aggregated, anonymized, and categorized data on search queries entered through Google’s search engine. Every day, billions of searches are conducted globally, and it would be computationally infeasible to process each one individually for public statistics. To make the data accessible in near real time, Google applies a random sampling procedure: it selects a representative subset of all searches, ensuring that the resulting data reflect the overall pattern of search activity without compromising user privacy.

Before publication, Google performs extensive data cleaning and aggregation. This process removes elements that could distort or endanger the representativeness of the sample. For example, searches containing personally identifiable information, queries with extremely low volume (such as a person’s full name), or repeated searches by the same user within a short time window. Through this filtering, the data are smoothed and anonymized, capturing general trends in collective search behavior rather than individual actions.

Once search counts are collected, Google normalizes them to make the data comparable across regions, time periods, and keywords. For a given search term k , location r , and time period t , Google first computes the relative frequency of that search term:

$$R_{k,r,t} = \frac{S_{k,r,t}}{S_{r,t}}, \quad (1)$$

where $S_{k,r,t}$ denotes the number of Google searches for term k in region r during period t , and $S_{r,t}$ is the total number of all searches in the same region and period. To facilitate interpretation, these ratios are then rescaled to a 0–100 index within the chosen time frame, where 100 corresponds to the period with the highest relative search intensity:

$$I_{k,r,t} = 100 \times \frac{R_{k,r,t}}{\max_t(R_{k,r,t})}. \quad (2)$$

Here $I_{k,r,t}$ represents the normalized Google Trends index, where 100 corresponds to the period of maximum relative search interest. All other observations are expressed as percentages of this maximum, so an index value of 50 means that the search term was half as popular relative to total search activity as in its peak week. Because the normalization is conducted within each query request, absolute search volumes cannot be inferred. The data are relative and dimensionless, suitable for temporal comparison but not for cross-query volume analysis.

Google Trends was first introduced in May 2006 as a public tool for visualizing search

interest and later replaced its predecessor, Google Insights for Search, in 2012. The underlying data, however, extend back to January 2004 and are updated continuously. Users can customize their data requests along several dimensions, including the time range (from a single day to the full history since 2004), the geographic scope (global, national, or subnational regions such as the German Bundesländer), and the thematic category (for example, “Jobs & Education”). Furthermore, users can specify the search type, such as Web, Image, News, Shopping, or YouTube searches, and disambiguate keywords through the “Topic” feature, which groups synonyms, misspellings, and semantically related expressions. For instance, searches for Apple Inc. can be distinguished from those for the fruit “apple.” These settings allow researchers to customize the scope and interpretation of the data while limiting contamination from unrelated search behavior.

2.3 Google trends stitching

As mentioned above, Google Trends returns weekly series that are normalized within each query window. Since a single request can cover at most 5 years at weekly frequency, any horizon beyond that must be assembled from overlapping downloads. Because each window is internally scaled (0–100) relative to its own maximum, raw levels are not comparable across windows; they must be re-scaled in the overlap before concatenation. Zimmermann’s original application (2004–2009) did not encounter this problem to the same extent because the analysis window fit comfortably within a single normalization regime.

Let $x_t^{(j)}$ denote the Google Trends index for a fixed search term k returned from window j on week t , with $j = 1, \dots, J$ ordered from oldest to newest. Suppose windows j and $j+1$ overlap on the set of weekly dates $\mathcal{O}_j$. Because $x_t^{(j)}$ and $x_t^{(j+1)}$ are each normalized internally, I estimate a scalar link factor that maps the scale of window $j+1$ onto window j . A robust choice uses the median ratio over the overlap, excluding zeros and non-finite values:

$$s_{j+1 \leftarrow j} = \text{median} \left\{ \frac{x_t^{(j)}}{x_t^{(j+1)}} : t \in \mathcal{O}_j, x_t^{(j)} > 0, x_t^{(j+1)} > 0 \right\}. \quad (3)$$

I then re-scale all observations in the later window by this factor,

$$\tilde{x}_t^{(j+1)} = s_{j+1 \leftarrow j} \cdot x_t^{(j+1)} \quad \text{for all } t \text{ in window } j+1, \quad (4)$$

and proceed sequentially from the oldest to the newest window. With three overlapping windows W_1, W_2, W_3 the chain-linking reads

$$s_{2 \leftarrow 1} = \text{median} \left(\frac{x_t^{(1)}}{x_t^{(2)}} \right)_{t \in \mathcal{O}_1}, \quad \tilde{x}^{(2)} = s_{2 \leftarrow 1} x^{(2)}, \quad s_{3 \leftarrow 2} = \text{median} \left(\frac{\tilde{x}_t^{(2)}}{x_t^{(3)}} \right)_{t \in \mathcal{O}_2}, \quad \tilde{x}^{(3)} = s_{3 \leftarrow 2} x^{(3)}. \quad (5)$$

A single weekly series $\{\tilde{x}_t\}$ is obtained by taking all dates from the earliest window and appending the non-overlapping tails of the re-scaled later windows, sorted on a strict 7-day (Sunday) grid. The median ratio in (3) is deliberately chosen for robustness: weekly overlaps may include zeros (clipping) and spikes (weeks set to 100 by within-window normalization), for which the mean ratio would be unduly sensitive.

A separate source of variation arises because Google delivers indices from daily random samples of the underlying query stream; identical requests on different days can yield slightly different series. A common remedy in the literature is to repeat the exact same download K times and average across pulls to reduce sampling noise (**CarriereSwallowLabbe2013**). Given the seminar timeline, I did not implement this pulling protocol here; I discuss it as a robustness extension and note that it is complementary to the stitching approach above. Two practical limitations merit acknowledgement. First, the link factors $s_{j+1 \leftarrow j}$ may be sensitive to structural breaks inside the overlap (for example, sudden spikes in attention or platform changes). I therefore exclude zero weeks, check the stability of s across sub-overlaps, and verify that results are qualitatively unchanged when using mean ratios or trimmed medians. Second, the stitched index remains a *relative share* of total Google activity for a given term; it is coherent over time *within* that term but not an absolute level, and indices from unrelated terms are not directly comparable unless retrieved jointly under identical parameters.

2.4 biweekly aggregation approach

The construction of biweekly indices follows the timing rationale of **Zimmermann2009**, adapted to the reporting cycle of German labor-market statistics. The Federal Employment Agency (Bundesagentur für Arbeit) releases unemployment figures around the end of each month at 10:00 a.m., but these data refer to conditions in the previous calendar month.¹ This institutional lag implies that agents, such as job seekers, firms, and media, receive official labor-market information with roughly a two-week delay relative to the underlying economic situation. In turn, search behavior in Google may react both to *real* labor-market changes (prior to the official release) and to the *public dissemination* of these statistics (after the release). Distinguishing between these two windows is crucial for a nowcasting framework: it allows one to separate “anticipatory” search activity from “reactionary” search activity triggered by the news itself.

¹For example, the January 2023 unemployment rate was released on 31 January 2023, referring to labor-market conditions as of 12 January 2023; see Bundesagentur für Arbeit, Veröffentlichungstermine Arbeitsmarktdaten. This lag is broadly stable across years.

To capture these dynamics, I construct two biweekly Google Trends aggregates per calendar month M :

- **W12_M**: search activity during the *first half* of month M (days 1–15), representing information available *before* the official release for month $M-1$.
- **W34_{M-1}**: search activity during the *second half* of the previous month ($M-1$), which is aligned forward to M for prediction. This window captures how users react to the newly published unemployment data for $M-1$ and potentially update expectations for M .

Unlike **Zimmermann2009**, who proportionally re-split weekly indices to match day-of-month boundaries exactly, I assign each week in its entirety to the half-month containing its Sunday timestamp. This “whole-week assignment” preserves the original Google Trends scaling and avoids distributing a single weekly value across two subperiods.

Formally, let $\{\tilde{x}_t\}_{t \in \mathcal{T}}$ denote the stitched weekly index for a given search term, observed at Sundays t with calendar date $(y(t), m(t), d(t))$. Define the weekly subsets

$$\mathcal{W}_M^{12} = \{ t \in \mathcal{T} : m(t) = M, d(t) \leq 15 \}, \quad (6)$$

$$\mathcal{W}_M^{34, \text{prev}} = \{ t \in \mathcal{T} : m(t) = M-1, d(t) > 15 \}, \quad (7)$$

where $\mathcal{W}_M^{34, \text{prev}}$ is shifted forward to align with month M . The biweekly aggregates are defined as simple averages:

$$X_M^{12} = \frac{1}{|\mathcal{W}_M^{12}|} \sum_{t \in \mathcal{W}_M^{12}} \tilde{x}_t, \quad X_M^{34, \text{prev}} = \frac{1}{|\mathcal{W}_M^{34, \text{prev}}|} \sum_{t \in \mathcal{W}_M^{34, \text{prev}}} \tilde{x}_t. \quad (8)$$

A concrete calendar illustration (without the three-Sundays edge case) is provided in Appendix 8.1. Because Sundays do not necessarily coincide with the 1st, 15th, or 16th of a month, the sets $\mathcal{W}_M^{12}$ and $\mathcal{W}_M^{34, \text{prev}}$ may contain one, two, or three observations. This design preserves the temporal coherence of weekly data while still aligning with the monthly unemployment release cycle.

The resulting biweekly panel spans June 2011 to January 2023, yielding $T = 140$ monthly observations. This horizon provides sufficient degrees of freedom for time-series estimation while allowing a clear separation between the information available *before* and *after* each official labor-market release. Consequently, the specification aligns naturally with a now-casting framework that tests whether search intensity during $W12_M$ can anticipate changes in unemployment, whereas $W34_{M-1}$ captures adjustments following the publication of the previous month’s data.

2.5 The selection of google trends variables

The selection of search indicators reflects how German job search is organized in 2011–2023 and is deliberately updated relative to **Zimmermann2009**. Whereas **Zimmermann2009** use four groups: (k1) *Arbeitsamt/Arbeitsagentur*, (k2) *Arbeitslosenquote*, (k3) *Personalberater/Personalberatung*, and (k4) a composite of major job boards, my study focuses on a smaller, behaviorally clearer set tied to modern search and application practices (2011–2023). This change tries to avoid news-driven terms and brands whose relevance changed markedly after 2010, while retaining the original idea of measuring inflows into unemployment. The first variable is the german word *Arbeitsamt* which shows the contact with the unemployment office. The idea is that it should capture information seeking around the public employment service, similar to k1 in **Askitas2009**. It is clearly retained as it proxies administrative touchpoints preceding with unemployment. The second one is *Jobbörse* which is the official job board usage. This relates to the public portal of the federal employment agency which is stable and policy relevant. It is as well part of Zimmermann's k4 set as it reflects search on a non commercial platform tightly linked to unemployment services. Moreover *stepstone* and *Indeed* which heavily relate to the private job board search intensity both are dominant private platforms in Germany over 2011–2023. We include these to capture the rest of **Askitas2009** broad k4 basket and to avoid brand drift. Finally *Bewerbug* is chosen, which measured the German application and CV activity and showcases the intent to apply. I believe it directly targets the application step and entails the importance of words like cover letters, templates and how to write and application. It can be seen preferably close to *Personalberater/Personalberatung* as of **Askitas2009** paper, which is niche, two-sided and changing in meaning over time. The two exclusions are intentional. First, *Arbeitslosenquote* (Zimmermann's k2) is in my belief primarily news-driven and risks hence capturing reactions to press releases rather than job search behavior itself. In a future nowcasting setup this could as well blur the distinction between pre-release information ($W34_{M-1}$, $W12_M$) and post-announcement reactions. Second the broad brand basket they formulated as their k4 is less robust in 2011–2023 because platform composition and market shares changed: consolidations, rebrandings, and varying SEO can inject artificial breaks. Narrowing to *Stepstone* and *Indeed* yields cleaner, conceptually stable proxies for private job-search intensity while *Jobbörse* preserves the public-portal channel.

2.6 Things to keep in mind

Before edging on to the explanatory analysis it is important to mention a few things. First as **Askitas2009** mentioned in his paper, Google Trends has been modifying and changing the way they collect data and the way we access the data. For instance it is known that Google Trends actually performs normalization on the data that at least exceed a certain

trashhold especially when it comes to the number of searches. The exact threshold though is not listed and hence the inclusion of other variables or the similarity when it comes to words and the procedure for forecasting or other analysis is not always given. As well, **Askitas2009** mention their approach that resplitts the activity proportionally in order to match the weekly base of google trends and that is necessary to perform their analysis. Because of this, it is possible that three weekly observations fall into one biweekly segment. However, the aggregation criterion remains internally coherent, and any small biases introduced by uneven week counts are expected to average out over the full twelve-year sample.

Even though as mentioned before google tried to disambiguate keywords through the topic feature Varian here claims that the probabilistic assignment google uses in order to shape its topics introduces further noise. He clearly emphasizes how a search query is partially assigned to all its relevant categories. He here clearly shows how apple could be weightedly attributed to information systems but at the same time as well to food and the weights are determined by google. As a consequence, this can lead to a bias during the classification process with regard to their interpretation. A researcher that might be interested in computer systems or information systems could find words such as apples that are inflated by people's interest in the food and with that in a different meaning.

Furthermore it is once again important to mention the in Carriers paper upcoming and detecting sample noise. With this, identical requests at different days return series that are similar but with minor changes. He here encourages a 24h sampling method that refetches the data every 24 hours. Once downloaded the data over a specific amount of days everyday, he suggests 50 days should be enough, the idea is to average out all the small differences. With this it is possible to reduce the measurement error that is introduced by google. The aspect is very related to the data quality of google.

Furthermore it is important to showcase Limitations that have been emphasized on, in Amuris paper. Like in other web search engines here it is important to mention that when using the google trend variables all the individuals seeking for job postings or for news with regard to the job market may well be not randomly selected among job seekers. Moreover, the indicators entail all job looking or job searching activities, which is the sum of all the searches performed by unemployed but at the same time by employed people. This can even more emphasize that while unemployed job search is believed to follow anti cyclical variation of job separation rates, on-the-job search is normally assumed to be cyclical. Here especially the keywords put in in the search engine do not always have to match people directly looking for jobs due to unemployment but can also cover students that are looking for working student jobs or internships and hence could have a short time impact on the labour market. So google just provides the normalized data but due to data privacy reasons do not provide a lot of information of which group of people especially when it comes to age or gender is actually in need of it and hence does not offer to give

more insight into more exploratory or subset analysis.

3 Deycriptive statistics and Visual Inspection

3.1 Descriptives

Before proceed with the unit-root and cointegration analysis, this section aims a structured look at the series. The aim here is concentrates towards (i) verifying that the biweekly aggregation is coherent, (ii) documenting broad comovement patterns between Google search indicators and unemployment, and (iii) motivating which simple correlations are informative for later ECM/VECM work. It will be interesting to see whether the W12/W34 design which has been extracted from Zimmerman shows any differences at first view as $W34_{M-1}$ capture the two weeks prior to release and $W12_M$ cover the two weeks after. This analysis hopes to gain additonal knowledge to check Zimmermans claim that the activities in $W34_{M-1}$ actually yield more predictive power than those in the later intervall $W12_M$, even though this period is closer to the announcement then the foremer (this is a cite).

The figures 1 (W12) and 2 ($W34_{\text{prev}}$) plot the unemployment rate alongside the biweekly Google indices. From the Figures several regularities emerge. First Public-portal terms (Arbeitsamt, Jobbörse) trend downward through 2019 and show a slight peak at 2020 broadly consitent with the decline in unemployment prior to the pandemic and the subsequent shock. Shorter-run movements come visibly with unemployment, also dring the Covid-19 period.

When it comes to private job-board terms (Indeed, Stepstone) we have a slightly different picture. Indeed exhibits a pronounced upward drift over the sample, reflecting the structural diffusion of the platform in Germany. Stepstone at first is more stationary-looking with some cyclical variability. Hence, as first thought for Indeed, a level correlation with unemployment can be misleading if the common variation is driven by trend rather than cycle.

The Bewerbung series displays marked seasonality (regular peaks in typical application months) and only a weak trend, which is consistent with application timing rather than sharp cyclical responses.

When comparing W12 with $W34_{\text{prev}}$, we could say taht at the medium-run horizon, W12 and $W34_{\text{prev}}$ convey very similar patterns, across all indicators, as the same local turning points and the same broad co-movement with unemployment are visible in both windows. The main difference is the volatility. $W34_{\text{prev}}$ shows more frequent and larger weekly spikes. This is expected for three reasons: (i) the $W34_{\text{prev}}$ window sits immediately before the monthly press release, a period with elevated media reporting and expectation pieces that can briefly elevate search activity even before the number is known. (ii) weekly Google indices are normalized within finite windows, so local maxima are clipped at 100

and can create spike-like features; (iii) our whole-week assignment to half-months means some month boundaries admit two or three weekly observations in the same subperiod, amplifying visible week-to-week variation. Overall, the variability pattern support using level with deterministic terms to accommodate structural drifts and, in robustness checks to purge seasonality.

3.2 Correlations

Because several search series contain trends and strong seasonality, we report three complementary correlation views that are informative yet conservative for time-series work. The contemporaneous correlation in levels which captures whether search intensity co-moves with unemployment at the monthly horizon showcases in our sample public-portal series (Arbeitsamt, Jobbörse) display relatively high positive correlations (roughly 0.7–0.8), Stepstone is moderate (≈ 0.5 –0.6), and Indeed is strongly negative (≈ -0.7 to -0.8), this reflects its structural rise and association with hiring rather than separations (Figures 1–2). Patterns are similar in W12 and W34_{prev}.

A first glance at the lead correlations in levels (one month ahead (X_t, U_{t+1})), provides a first glance at potential nowcasting content. The ranking nevertheless remains similar public-portal terms retain positive lead correlations, Stepstone is modestly positive, and Indeed remains negative. Differences between W12 and W34_{prev} are small; if anything, W12 sometimes loads a bit more on contemporaneous reactions to the release, while W34_{prev} sometimes shows slightly higher variation due to pre-release attention spikes. These differences are not large enough to be diagnostic by themselves.

Finally the Contemporaneous correlation in year-over-year differences $((\Delta_{12}X_t, \Delta_{12}U_t))$. This removes seasonal patterns and most deterministic drift. Results here remain positive for public portal series and weaker for Stepstone. Indeed’s correlation becomes closer to zero (or modestly negative), consistent with the notion that its strong level trend is structural rather than cyclical. This view is the safest descriptive statistic to report before ECM/VECM, because it is least prone to spurious correlation from trends.

We intentionally do not over-interpret small differences between W12 and W34_{prev} in these correlations. The descriptive evidence indicates broadly similar information content across windows, with W34_{prev} exhibiting more high-frequency noise. Whether one window materially outperforms the other is therefore an empirical question to be settled in the Johansen/ECM framework with out-of-sample evaluation.

3.3 Interim takeaways for the econometric setup

First of all, both W12 and W34_{prev} track the medium-run path of unemployment; the main difference is that W34_{prev} is noisier with more spikes, plausibly due to pre-release media attention, Google’s within-window normalization, and whole-week assignment at

month boundaries. The public portal terms are procyclical; private-board terms are heterogeneous: Indeed is dominated by structural adoption (negative level correlation with unemployment), while Stepstone shows clearer cyclical variation. In order to protect against spurious correlation, subsequent models should allow for deterministic components (linear trend or breaks where appropriate) and verify stationarity on differences (ADF), followed by cointegration testing (Johansen) and an ECM/VECM that admits distinct short- and long-run channels.²

4 The Augmented Dickey Fuller test

In order to later test for cointegration relationships using the Johansen Trace test and in case of finding them modeling with an Vector error correction model we need to perform the Augmented Dickey Fuller test, in order to see whether the series is nonstationary in levels and exhibits an unit root but is stationary in first differences. It is important to mention here that the plain Dickey Fuller test considers the case of an AR(1). Nevertheless as we here have macroeconomic data especially the unemployment rate it is common that the data's residuals are serially correlated. The Augmented Dickey Fuller test fixes this nevertheless by adding lagged differences. .

The ADF regression with p augmenting lags and deterministic component d_t is

$$\Delta y_t = \alpha + \delta t + \phi y_{t-1} + \sum_{i=1}^p \psi_i \Delta y_{t-i} + \varepsilon_t, \quad (9)$$

where Δ denotes the first difference and with that the number of lags included in the model. For the test purpose the Lag length in the ADF tests is selected by AIC with an initial upper bound of 12 which is to be seen standard for monthly data, as it captures yearly autocorrelation. The critical values are used from MacKinnon (2010) as implemented in statsmodels. Important is that I differentiate the respective different Augmented Dickey Fuller test by eyeballing at the series. For every respective Dickey Fuller test hence (α, δ) are included or excluded according to the deterministic specification. The purpose of this individual alignment is that it is necessary to choose the correct testing case in order to obtain enough power in order to being able to reject a false null hypothesis. The test case needs to be chosen correctly in order that the case has enough power to actually detect whether there might be a constant or even a constant and a drift in the series.

Hence I test :

- $H_0: \phi = 1$ (unit root) against
- $H_1: \phi < 0$

²This mirrors the spirit of ? but replaces their implicit assumptions on integration/cointegration with explicit pre-testing and a Johansen/VECM specification.

As mentioned there are three cases considered for the unit root tests in levels: (i) **nc** (no constant, no trend: $\alpha = \delta = 0$), (ii) **c** (drift: $\delta = 0$), and (iii) **ct** (drift+trend). The results of the Augmented Dickey Fuller test for the series in levels can be seen in Table 4. The cases for the respective series are mainly chosen by eyeballing. For the stepstone variable we choose for the weeks12 the constant case as it looks like the series is fluctuating over a nonzero mean and the previous stepstone series looks similar as for this reason I choose the test scenario. Indeed time series on the other hand exhibits an upward moving trend in the w12 version but as well in the previous w34 version. For this reason I will be choosing include the trend case to have enough power for this test. For jobbörse we see a similar pattern as here both series exhibit downward moving trends which as a consequence I choose the trend case for those two series. Arbeitsamt equally as Jobbörse as well exhibit a downward trend which is why I am choosing the trend case. Even if in the Bewerbung W12 variable there is this strong seasonal pattern I will as well for this variable choose the trend case as well as for its W34prev counterpart. The downward moving trend is clearly visible. Finally for the unemployment variable first it appears that a downward moving trend is observable but on second sight the series is clearly moving up for what I will choose the constant case. Looking at the critical values and in this connection the p values of the series in table 4 we cannot reject the null hypothesis of a unit root for the series. That's why on a second glimpse we will take a look at the first differences of the data to see how they look like. For the first differences we will choose the first case of the Augmented Dickey Fuller test. Even though in practice seldomly used that case pictures the behaviour of our first difference data. The first differences of our series always show the behaviour of fluctuating over a zero mean. Furthermore I calculated the mean of all the first difference series and conducted a t-test in order to check whether the non constant case is justified. All the p values of the means in levels are extremely high. That's why I can reject the null hypothesis of mean = 0. That's why the series clearly fluctuate around a nonzero mean. So even though the case is rarely used in practice due to the power of this test, I believe that my security checks and the eyeballing seem convincing enough to make an case one of the Augmented Dickey Fuller test. The results shown in table 5 clearly here show how all variables are stationary in first differences and hence we can move over to check for cointegrating relationships.

5 Tables

5.1 Tables used by Zimmerman

Table 1: Google Trends variables used and rationale (Germany, Web Search, 2011–2023)

Code	Search Terms / Topics	Intended Interpretation
k1	<i>Arbeitsamt, Arbeitsagentur</i>	Contact with unemployment office; proxy for inflows into registered unemployment.
k2	<i>Arbeitslosenquote</i>	News-driven attention to unemployment statistics; measures public awareness rather than direct search.
k3	<i>Personalberater, Personalberatung</i>	Job-placement and recruitment services; indicates professional job search or hiring activity.
k4	<i>Stepstone, Monster, Jobscout, Meinstadt, Jobbörse</i>	Major job-board brands (public and private) capturing general online job-search intensity.

5.2 Tables used by me

Variable	Search (Google Trends)	Topic	Behavioral Interpretation
Arbeitsamt	“Arbeitsamt” (topic)		Information seeking related to public employment offices; inflow indicator.
Jobbörse	“Jobbörse” (Bundesarbeitsagentur für Arbeit)		Public job-board use, aligned with official labor-market services.
Stepstone	“Stepstone” (topic)		Major private job portal; stable representation of private job-search behavior.
Indeed	“Indeed” (topic)		Meta-search aggregator for job listings; reflects modern, consolidated online job-search practice.
Bewerbung	“Bewerbung” (topic)		Search activity related to preparing applications or CVs; measures intention to apply.

5.3 Table Mean and Standard

Table 2: Summary of Mean and Standard Deviation for Key Variables

Variable	Mean	Std. Dev.
indeed_W12	87.77	39.97
indeed_W34prev	84.64	39.47
jobbörse_W12	52.13	24.17
jobbörse_W34prev	50.27	23.82
arbeitsamt_W34prev	44.30	16.02
arbeitsamt_W12	43.89	15.67
stepstone_W34prev	61.71	14.01
stepstone_W12	63.68	13.00
bewerbung_W34prev	55.66	14.36
bewerbung_W12	58.56	13.83
Unemp	6.00	0.72

5.4 Table correlations

Table 3: Lead correlations with next month's unemployment rate

Variable	Corr(Variable, Unemp _{t+1})
stepstone_W12	0.55
indeed_W12	-0.78
jobbörse_W12	0.73
arbeitsamt_W12	0.72
bewerbung_W12	0.59
stepstone_W34prev	0.45
indeed_W34prev	-0.78
jobbörse_W34prev	0.68
arbeitsamt_W34prev	0.64
bewerbung_W34prev	0.47

Table 4: Augmented Dickey–Fuller tests in levels (monthly data, 2011M6–2023M1)

Series	Det. spec	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%?
stepstone_W12	c	12	-1.4933	-2.8846	0.5368	No
indeed_W12	ct	12	-1.8816	-3.4456	0.6641	No
jobbörse_W12	ct	12	-3.1492	-3.4456	0.0950	No
arbeitsamt_W12	ct	11	-1.7611	-3.4454	0.7232	No
bewerbung_W12	ct	12	-2.0718	-3.4456	0.5619	No
stepstone_W34prev	c	12	-1.2275	-2.8846	0.6616	No
indeed_W34prev	ct	12	-1.4060	-3.4456	0.8592	No
jobbörse_W34prev	ct	12	-3.0425	-3.4456	0.1206	No
arbeitsamt_W34prev	ct	12	-2.0211	-3.4456	0.5899	No
bewerbung_W34prev	ct	12	-1.9437	-3.4456	0.6317	No
Unemp	c	12	-2.2101	-2.8846	0.2026	No

Table 5: ADF tests for first differences (regression: no constant, monthly data 2011M6–2023M1)

Series	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%?
stepstone_W12	11	-3.2242	-1.9433	0.0013	Yes
indeed_W12	12	-2.4965	-1.9433	0.0121	Yes
jobbörse_W12	11	-3.0608	-1.9433	0.0022	Yes
arbeitsamt_W12	10	-5.5334	-1.9433	0.0000	Yes
bewerbung_W12	12	-3.4875	-1.9433	0.0005	Yes
stepstone_W34prev	11	-4.0136	-1.9433	0.0001	Yes
indeed_W34prev	12	-2.2670	-1.9433	0.0225	Yes
jobbörse_W34prev	11	-2.9936	-1.9433	0.0027	Yes
arbeitsamt_W34prev	12	-3.6222	-1.9433	0.0003	Yes
bewerbung_W34prev	11	-5.1217	-1.9433	0.0000	Yes
Unemp	12	-2.3236	-1.9433	0.0194	Yes

5.5 Table DF levels

5.6 Table first differences ADF

6 Application and Results

6.1 Robustness and Sensitivity

7 Figures

7.1 Biweekly aggregation

Figure 4 illustrates the assignment on a Sunday grid: Sundays falling on or before the 15th are collected into $W12_M$; Sundays after the 15th of $M-1$ are collected into $W34_{M-1}$ and aligned to M .

7.2 Week w12

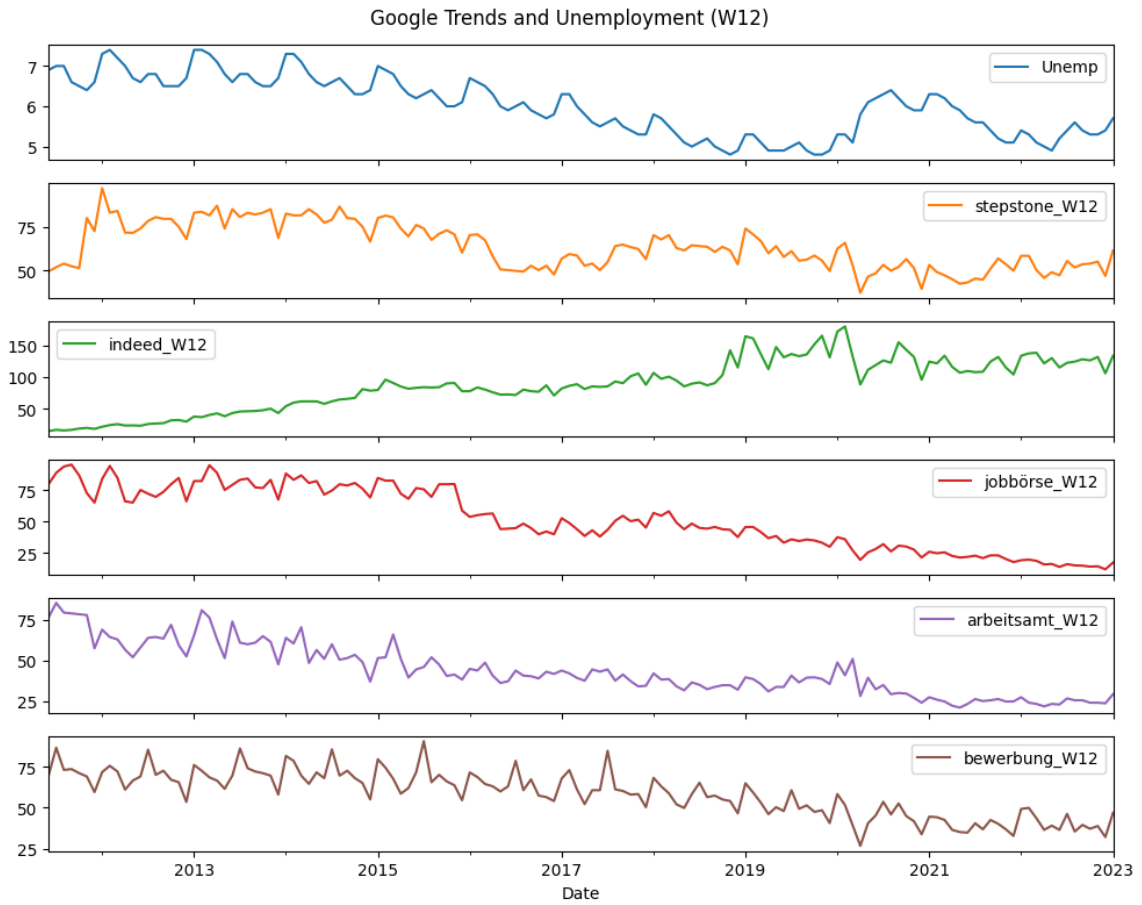


Figure 1: Monthly Unemployment and Google Search Indicators (W12, 2011–2023)

7.3 Week w34



Figure 2: Monthly Unemployment and Google Search Indicators (W34prev, 2011–2023)

7.4 Correlations

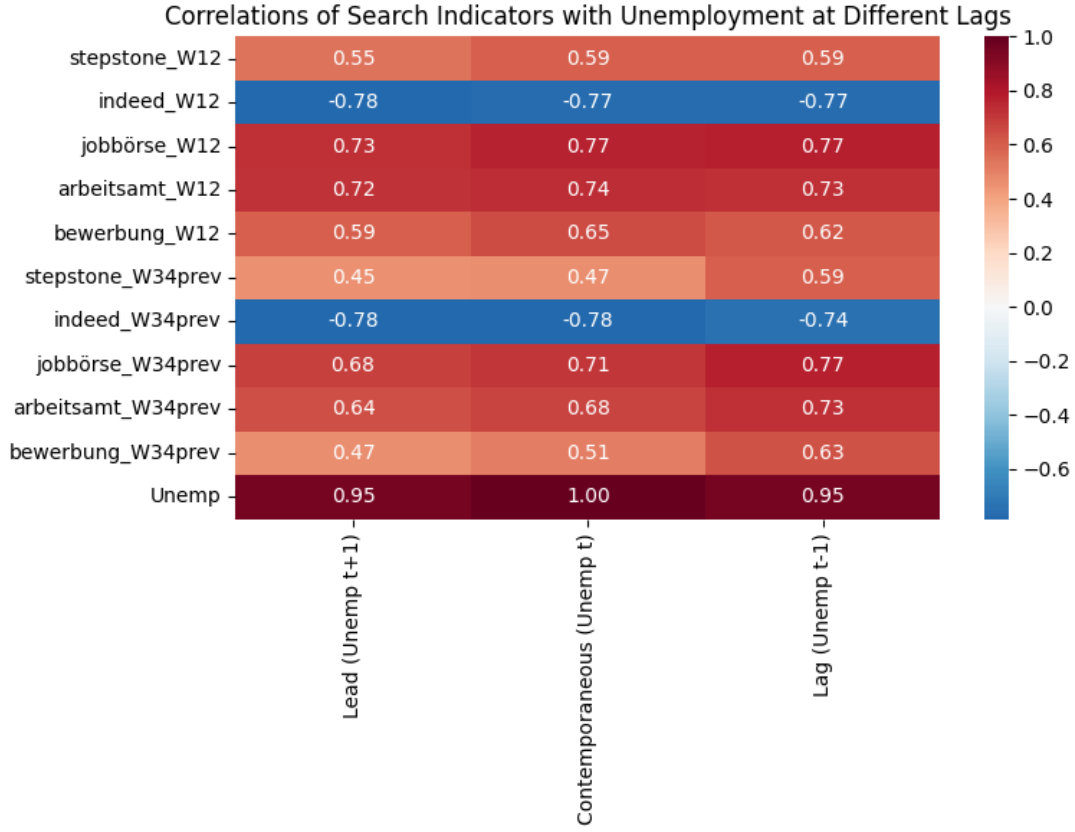


Figure 3: Lead correlations of Google search indicators with next month’s unemployment rate. Positive values indicate anticipatory search behavior.

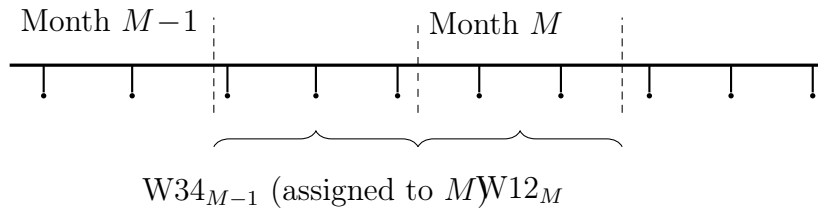


Figure 4: Assignment of weekly (Sunday) observations to W34 $_{M-1}$ and W12 $_M$ on a Sunday grid. Dashed lines indicate the 15th of each month; the month boundary is shown by the vertical solid line.

8 Appendix

8.1 Calendar example for the biweekly assignment

This appendix provides a concrete example of how weekly (Sunday) observations are assigned to the biweekly aggregates W12 $_M$ and W34 $_{M-1}$ using the “whole-week” rule described in the main text. The example avoids the three-Sundays edge case.

Consider $M = \text{June } 2022$. The relevant Sundays around this month are:

$\{ 2022-05-15, 2022-05-22, 2022-05-29, 2022-06-05, 2022-06-12, 2022-06-19, 2022-06-26 \}$.

The assignment follows the definitions in Eqs. (6)–(7):

$$\mathcal{W}_{\text{Jun}22}^{12} = \{ 2022-06-05, 2022-06-12 \}, \quad \mathcal{W}_{\text{Jun}22}^{34,\text{prev}} = \{ 2022-05-22, 2022-05-29 \}.$$

Here, Sundays in the first half of June (days ≤ 15) enter $\text{W12}_{\text{Jun}22}$. Sundays after the 15th of May (the second half of $M-1$) enter $\text{W34}_{\text{May}22}$, which is aligned forward to June for prediction.

For a stitched weekly index $\{\tilde{x}_t\}$, the biweekly aggregates used in Eq. (8) evaluate to simple two-point averages:

$$\begin{aligned} X_{\text{Jun}22}^{12} &= \frac{1}{2} \left(\tilde{x}_{2022-06-05} + \tilde{x}_{2022-06-12} \right), \\ X_{\text{Jun}22}^{34,\text{prev}} &= \frac{1}{2} \left(\tilde{x}_{2022-05-22} + \tilde{x}_{2022-05-29} \right). \end{aligned}$$

For completeness, the Sundays 2022-06-19 and 2022-06-26 belong to the *second* half of June and therefore contribute to $\text{W34}_{\text{Jun}22}$, which is aligned to July ($M+1$) in the main biweekly panel.

8.2 Illustration of Google Trends normalization

Google Trends returns search indices scaled from 0 to 100 within the chosen time window and parameter set. For instance, suppose two terms—“Stepstone” and “Indeed”—are queried separately for the same period and region. If the true underlying search volumes are in the ratio 5:1, the normalized indices might still show peaks at 100 for both, since each is internally rescaled to its own maximum. Thus, a value of 50 for “Stepstone” and 50 for “Indeed” does not imply equal absolute search interest but indicates that each term reached half of its own peak intensity within the selected window. Because of this internal normalization, absolute comparisons across search terms are not meaningful unless the terms are retrieved jointly in one query.

9 safes

9.1 This I still mention in intro

Carrier: CARRIER EXTRA Focussing on automobile sales in Chile they constructed a Google Trends Automotive Index (GTAI) from brand-related search queries and incorporated it into simple autoregressive nowcasting models. Their results showed that

models including search data significantly improved both in-sample and out-of-sample forecast accuracy and better captured turning points in sales dynamics. Moreover using Granger Causality they confirm that search activity contains information that precedes official sales data, rather than merely reacting to it. They further examined whether these search indices could anticipate future developments and found that the predictive content of Google Trends is not limited to contemporaneous relationships but also provides forward-looking signals of economic activity.

Goel: Using Yahoo! search data, they analyzed over 300 entertainment products, including movie releases, video game sales, and Billboard music rankings to test whether search intensity could predict real-world outcomes. Their results showed that search queries were highly correlated with future demand, forecasting box-office revenues, video game sales, and song popularity days or even weeks before their release. Their study emphasized forecasting forward-looking behavior rather than contemporaneous tracking, and benchmarked search-based predictions against models using traditional data (e.g., production budgets, critic ratings, and prediction market data). They found that search activity added most predictive value when conventional data were scarce, demonstrating that web queries can complement or even substitute for established information sources in real time. Importantly, Goel et al. highlighted that search data’s utility depends on context. The predictive advantage is strongest for new or unstructured markets (such as new video games) and weaker when detailed baseline data already exist.

A Additional Material